8 **(a)** State what is meant by a *field of force*.

.....

..... [1]

(b) Fig. 8.1 shows the setup of a high-resolution spectrometer within a vacuum. It is used to measure masses of singly positive charged ions (+*e*) which are released from an ion source.

The ions enter a velocity selector at right angles to both electric (*E*) and magnetic (B_1) fields. The uniform electric field (*E*) is setup with a pair of horizontally charged plates, namely plate P and plate Q. The uniform magnetic flux density B_1 is directed into the plane of the paper.

Only ions with velocity *v* pass through the velocity selector undeflected, before entering into another uniform magnetic field (B_2) region. The ions X and ions Y follow semicircular paths before reaching a detection surface as shown in Fig. 8.1.

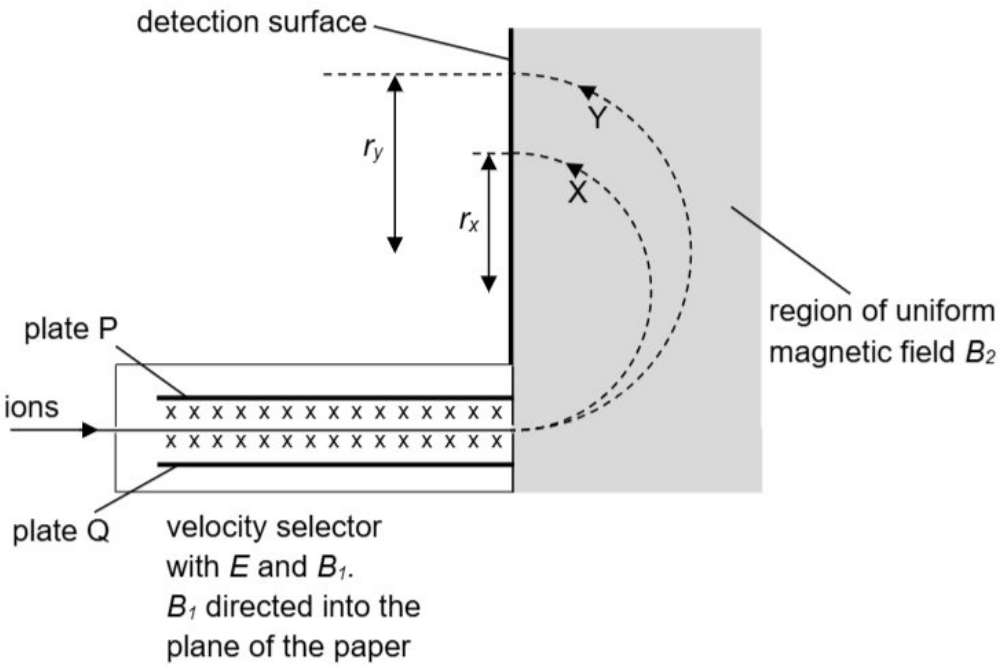


Fig. 8.1

The measured electric field strength, flux densities, radii of the paths of the ions and their associated uncertainties are shown below.

electric field strength *E* / V m^{-1}	$3.0 \times 10^4 \pm 2\%$
magnetic flux density B_1 / T	$0.25 \pm 1\%$
magnetic flux density B_2 / T	$0.60 \pm 1\%$
radius of path of ion *X*, r_x / m	0.074 ± 0.001
radius of path of ion *Y*, r_y / m	0.079 ± 0.001

(i) Explain why the ions pass undeflected through the velocity selector.

.....

.....

..... [2]

(ii) State and explain which plate, P or Q is at a higher potential.

.....

.....

..... [2]

(iii) Calculate the ions' velocity v.

v = ... m s^{-1} [2]

(iv) State and explain the direction of the magnetic flux density B_2.

.....

.....

.....

..... [2]

(v) Explain whether there is work done on the ion as it moves within B_2.

.....

.....

.....

..... [2]

(vi) Considering the path of ion within B_2 region, show that the expression for the ion mass is $m = \frac{eB_1B_2r}{E}$.

where m is the ion mass and r is the radius of the semicircular path by the ion.

Explain your working.

[2]

(vii) Calculate the mass of ion X.

mass of ion X = ... kg [1]

(viii) Calculate the percentage uncertainty of the mass of ion X.

percentage uncertainty = ... % [2]

(ix) The spectrometer resolution criterion is given by the equation

$$\Delta r > 3\sigma$$

where Δr is the difference in the radius of path of the two X and Y ions and σ is the absolute uncertainty of the radius of the path.

Determine whether the ions are distinguishable.

……

…… [2]

(x) Suggest and explain one experiment modification to improve resolution without changing E.

……

……

…… ……

…… [2]